SIMATS ENGINEERING

CSA07 – Computer Networks

CO1: Apply the functions of OSI layers in enabling reliable data transmission across networks

Question 4

In the OSI model, reliable data transmission is achieved through the combined function of the Data Link Layer and the Transport Layer, both of which take part in error control. The Data Link Layer performs error detection at the local level, while the Transport Layer performs error correction at the end-to-end level. Together, they form a two-tier error control mechanism that improves the overall reliability of network communication.

The Data Link Layer (Layer 2) is responsible for detecting errors within each frame transmitted across a single link between two directly connected devices. It uses techniques such as parity checking, checksum calculation, and Cyclic Redundancy Check (CRC) to verify whether the received frame matches the frame that was originally sent. If a frame is found to be corrupted, it is discarded immediately at that link, preventing damaged data from travelling further across the network.

The Transport Layer (Layer 4) is responsible for ensuring that data is delivered correctly from the source host to the destination host, regardless of how many intermediate links or devices the data passes through. It uses sequence numbers, acknowledgments (ACK/NACK), and retransmission timers to confirm that every segment has arrived intact. If a segment is lost or corrupted anywhere along the path, the Transport Layer requests retransmission from the original sender.

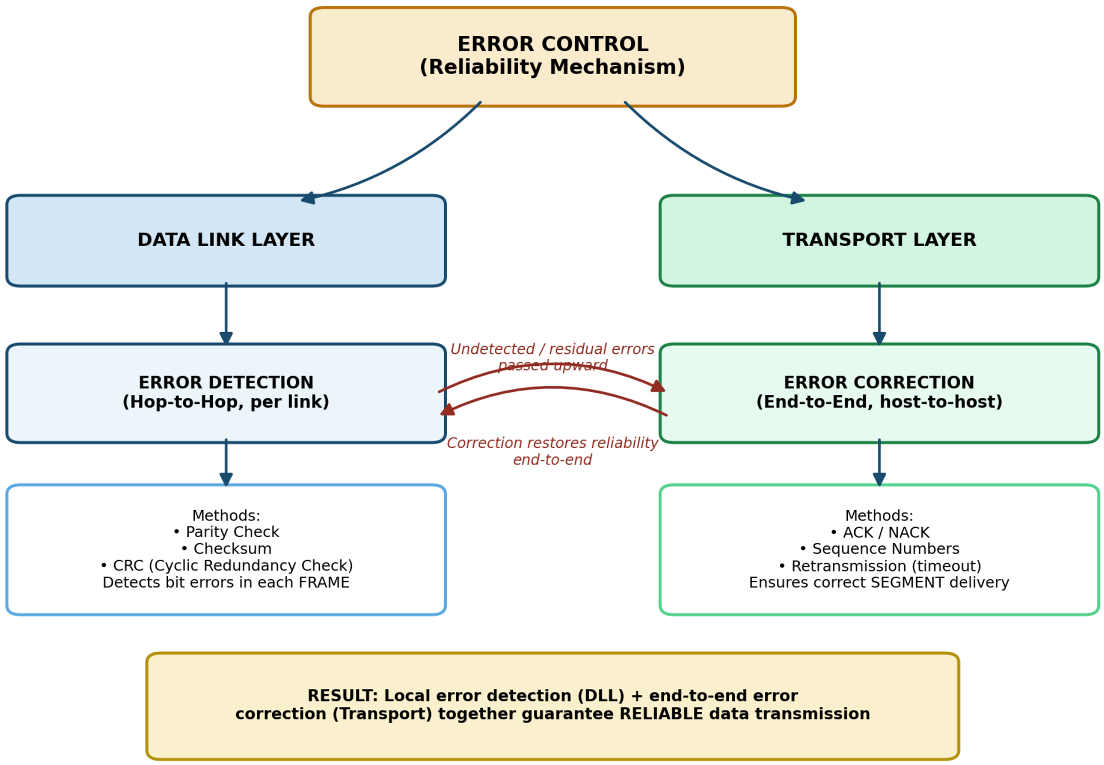
Analysis:

- The Data Link Layer only protects a single hop; it cannot detect an error that occurs later on a different link, nor can it confirm that data finally reached the destination.
- The Transport Layer closes this gap by operating end-to-end, verifying the complete path from sender to receiver.
- Detecting errors early at the Data Link Layer prevents corrupted frames from being carried further, saving network bandwidth.
- Correcting errors at the Transport Layer guarantees that the receiving application ultimately gets accurate and complete data, even if a lower-layer failure goes unnoticed at some point along the path.
- The interaction between these two layers therefore combines fast, local error detection with guaranteed, end-to-end error correction.

Conclusion

The Data Link Layer provides frame-level error detection using CRC and checksum techniques, while the Transport Layer provides end-to-end error correction through acknowledgments and retransmission. Their combined operation ensures that data transmitted across a network remains accurate and reliable, even when errors occur at intermediate links.

The concept map illustrating this relationship is shown below:



Question 5

When a user opens a webpage in a browser, the request and response travel through all seven layers of the OSI model, both on the client side and on the server side. Each layer performs a specific function that transforms the data and prepares it for transmission across the network.

The Application Layer generates the HTTP or HTTPS request, such as a GET request for a webpage. The Presentation Layer formats and encrypts this data, using protocols such as SSL/TLS to secure the communication and format the content correctly. The Session Layer establishes and maintains the connection session between the browser and the web server for the duration of the exchange.

The Transport Layer divides the HTTP data into segments and uses TCP to establish a reliable connection on port 80 or 443, ensuring that all segments arrive in the correct order. The Network Layer encapsulates these segments into packets and adds IP addressing information, allowing routers to determine the correct path to the destination server. The Data Link Layer converts these packets into frames and adds MAC addressing for delivery within the local network. Finally, the Physical Layer converts the frames into bits and transmits them as electrical, optical, or radio signals over the physical medium.

Analysis:

- If the Physical Layer fails, no signal is transmitted at all, and the request never leaves the device, resulting in a complete communication breakdown.
- If the Data Link Layer fails, frames cannot be delivered within the local network, so the device cannot even reach its router or gateway.

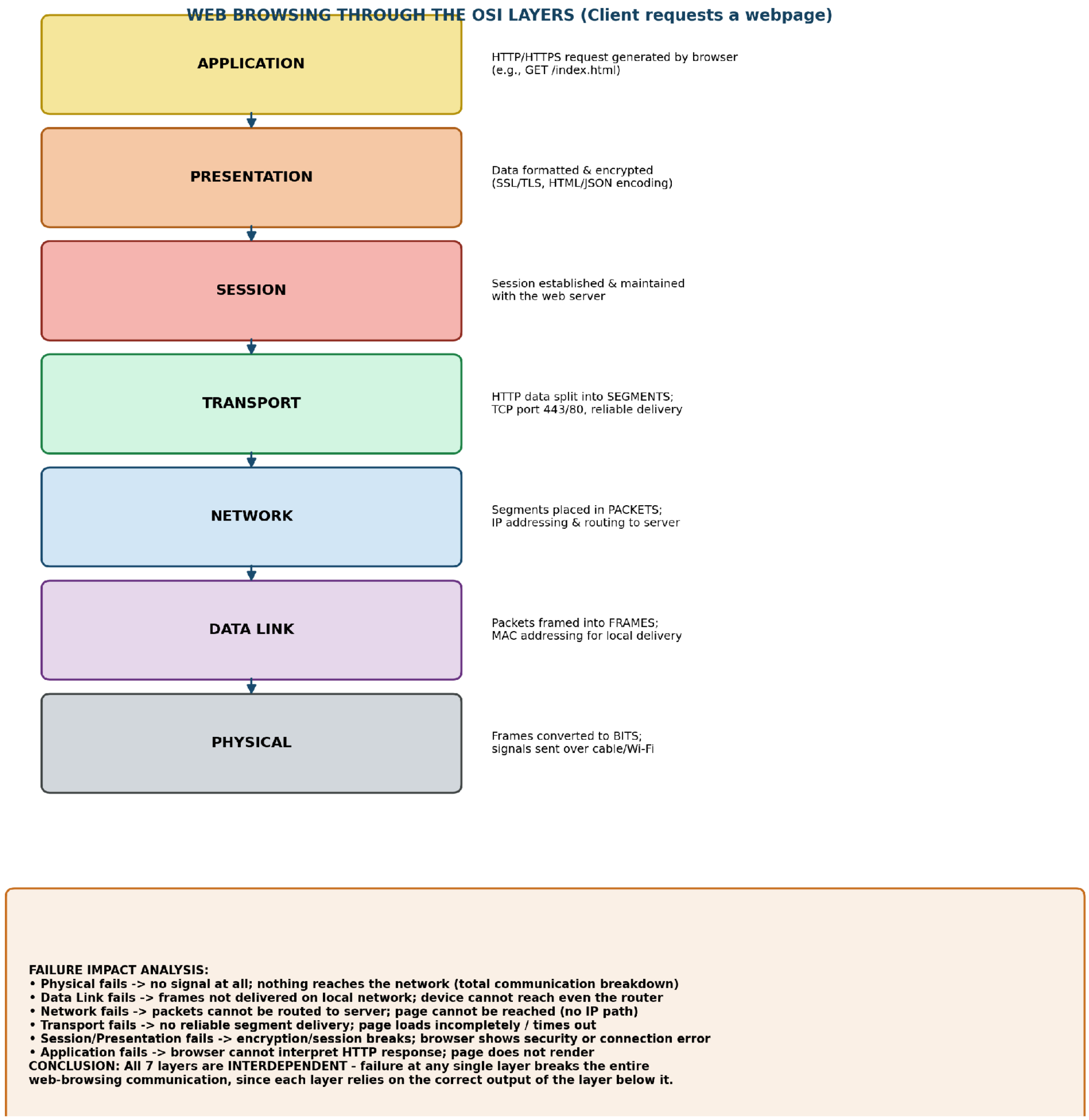
- If the Network Layer fails, packets cannot be routed to the correct IP address, making the website unreachable even though the local connection is working.
- If the Transport Layer fails, segments are not reliably delivered or reassembled, causing the page to load incompletely or time out.
- If the Session or Presentation Layer fails, the secure session or encryption breaks, and the browser displays a connection or certificate error.
- If the Application Layer fails, the browser cannot interpret the HTTP response, so the page does not render even if all the data has arrived correctly.

This shows that the seven layers are closely interdependent, since each layer relies completely on the correct functioning of the layer below it.

Conclusion

Web browsing requires the coordinated functioning of all seven OSI layers, from generating the HTTP request at the Application Layer to transmitting bits at the Physical Layer. A failure at any single layer disrupts the entire communication process, because there is no alternate path for data to bypass a layer that is not functioning correctly.

The concept map illustrating this flow is shown below:



Question 6

In OSI-based troubleshooting, each layer of the network is checked in sequence, starting from the Physical Layer and moving upward to the Application Layer, since every upper layer depends on the correct functioning of the layer beneath it. In the given concept map, the Physical Layer and Data Link Layer are both working correctly, while the Network Layer, Transport Layer, and Application Layer are all failing.

The fact that the Physical Layer is working confirms that the cabling, network interface card, and the electrical or optical signal are all functioning correctly. The fact that the Data Link Layer is working confirms that MAC addressing and framing are successful, meaning the device is able to communicate correctly within its local network.

The first failure appears at the Network Layer, which indicates a problem with IP addressing, subnetting, routing, or the default gateway configuration. Because packets cannot be forwarded correctly at this layer, they never reach their intended destination.

Analysis:

- Since the Transport Layer depends on the Network Layer to deliver packets, its failure is a direct consequence of the Network Layer problem, and not a separate fault of its own.
- Since the Application Layer depends on the Transport Layer to deliver segments, it also fails as a result, even though the application software itself may be working correctly.
- This means the single failure at the Network Layer cascades upward, causing both the Transport Layer and the Application Layer to fail as well.
- The root cause of the issue is therefore isolated to the Network Layer, most likely due to a misconfigured IP address, an incorrect subnet mask, or an unreachable gateway or router.

Conclusion

By mapping the status of each OSI layer from bottom to top, the concept map allows a technician to pinpoint exactly where communication breaks down instead of checking the entire system randomly. Since the Physical and Data Link Layers are confirmed to be working, troubleshooting effort can be directed specifically toward the Network Layer configuration, saving time and avoiding unnecessary checks on layers that are already functioning correctly.

The concept map illustrating this troubleshooting flow is shown below: